DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 4: The Expected Utility Hypothesis

Why Expected Value Isn't Enough · Risk Attitudes · Jensen's Inequality

August 3, 2026

Today's Agenda

1. Why expected monetary value fails: the St. Petersburg Paradox

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4. Jensen's Inequality

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- Use Jensen's Inequality.

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Heads: win Rs. 1000. Tails: lose Rs. 1000. Expected value: exactly Rs. 0. Yet almost everyone declines. Scale it up to $\pm$Rs. 10 crore: still $\mathbb{E}[W] = 0$ – but now refusal is nearly universal.

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Lottery: Definition

Definition

A **lottery** (gamble) is a probability distribution over monetary outcomes: payoffs $W_1, \ldots, W_n$ with probabilities $p_1, \ldots, p_n$ summing to 1.

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Infinite expected value – yet real people offer only a modest sum.

Resolving the Paradox

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Rs. 0 $\rightarrow$ Rs. 1000 feels transformative. Rs. 10,00,000 $\rightarrow$ Rs. 10,01,000 – the same Rs. 1000 – barely registers. Just like food: the first pizza slice when hungry is wonderful; the tenth, when full, adds little. Bernoulli: people maximize *expected utility* $\mathbb{E}[u(W)]$ using a **concave** $u(\cdot)$ – not $\mathbb{E}[W]$ itself.

The Expected Utility Hypothesis

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Not arbitrary

Von Neumann–Morgenstern (1944): 4 rationality axioms (completeness, transitivity, continuity, independence) $\Rightarrow$ preferences *must* be representable this way.

A Note on Units

Remark

$u(W)$ has no units, no natural zero.

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$u(W)$ has no units, no natural zero. Only *shape* and *relative* comparisons matter – you can't say "my utility is 4.7," only "I prefer A to B because $\mathbb{E}[u_A(W)] > \mathbb{E}[u_B(W)]$."

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Definition (for a fair gamble with mean $\overline{W}$)

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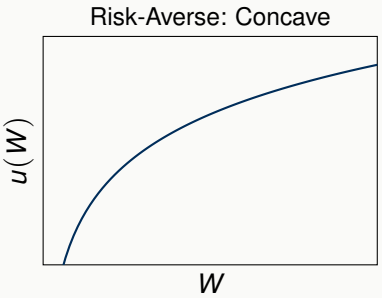
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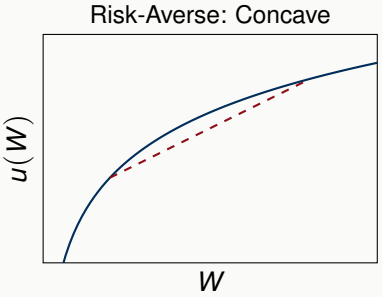
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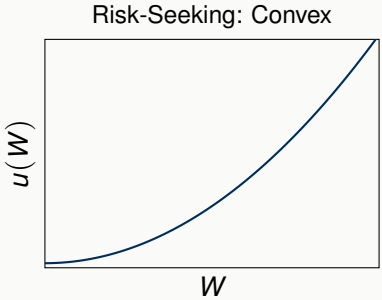
Concave $u \to$ risk-averse Linear $u \to$ risk-neutral Convex $u \to$ risk-seeking

Concave vs. Convex Utility

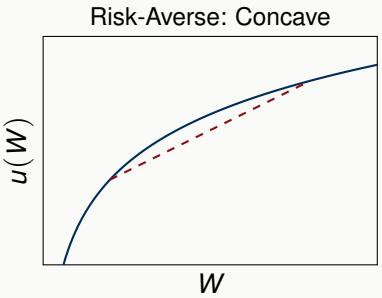


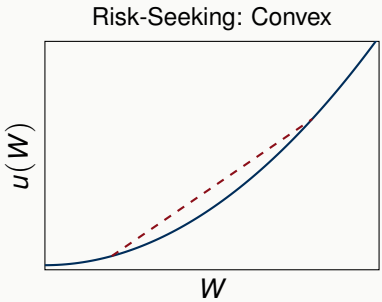
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Reading It Off the Picture

The curve (concave u) always sits *above* the chord connecting two points on it. The chord's midpoint height $= \mathbb{E}[u(W)]$ for a 50-50 gamble; the curve's height there $= u(\overline{W})$. Curve above chord $\Rightarrow u(\overline{W}) > \mathbb{E}[u(W)]$ – the entire content of risk aversion, in one picture.

Example: Coin Flip Under Log Utility

$u(W) = \ln(W)$, gamble: Rs. 50 or Rs. 200, equal prob.

$$\mathbb{E}[u(W)] = 0.5\ln(50) + 0.5\ln(200) = 0.5(3.912) + 0.5(5.298) = 4.605$$

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This gap is exactly what we'll call the risk premium – next lecture.

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- Curvature of $u \Leftrightarrow$ risk attitude: concave/linear/convex $\rightarrow$ averse/neutral/seeking.
- **Jensen's Inequality**: $\mathbb{E}[u(W)] \leq u(\mathbb{E}[W])$ for concave u – the mathematical heart of risk aversion.

Check Your Understanding

Discuss (2 minutes)

1. Lottery: Rs. 0 or Rs. 400, equal prob. Under $u(W) = \sqrt{W}$, verify Jensen's inequality.

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3. Would anyone buy insurance if u were linear? Why or why not?

Looking Ahead

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Making Risk Aversion Measurable

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Lecture 5: Certainty Equivalents & the Arrow–Pratt Coefficients